

Cool, Chill, and Cruel: Probabilistic Ranking with Greedy Approach for Recovering Sparse Binary Secrets in LWE

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Motivation & Overview

Why LWE?

- *Post-Quantum Cryptography*: LWE underpins many PQC proposals.
- *Homomorphic Encryption*: Real-world HE often relies on LWE for secure computations on encrypted data.

Sparse Binary Secrets:

- Highly efficient for certain HE operations (less overhead).
- But potentially more vulnerable to targeted cryptanalysis.

Problem:

- Partial lattice reduction can split columns into “hard” vs. “easy,” but real data sometimes shows a *slope* in variance.
- We want a refined approach to handle that intermediate region effectively.

Existing Approaches & Problem Statement

Learning With Errors (LWE) Recap:

- *Matrix form:* $\mathbf{b} \equiv \mathbf{A}\mathbf{s} + \mathbf{e} \pmod{q}$,
- $\mathbf{s} \in \{0, 1\}^n$ (sparse binary secrets with Hamming weight h),
- $\mathbf{e}$ is “small-norm” noise, drawn from a certain distribution.

Existing Attacks on LWE (small secrets):

- **Primal/Dual & Hybrid MiTM:** reduce dimensions or guess partial bits.
- **ML-based:** SALSA, PICANTE, VERDE use advanced classifiers to find the secret bits efficiently.
- **Cool & Cruel Approach:** partial lattice reduction $\rightarrow$ columns labeled as either high or low variance.

Problem Statement:

- Real data can show *gradual* variance transitions (a “slope”), not a clean hard/easy split.
- **Goal:** Extend beyond the two-zone (Cruel vs. Cool) partition to handle *medium-variance* columns more effectively.

Our Contribution

Three-Zone Partition:

- **Cruel region:** near-uniform variance $\approx q/\sqrt{12}$, brute force or meet-in-the-middle.
- **Chill region:** moderate variance “slope,” handled with probabilistic or ML-based partial enumeration.
- **Cool region:** fully reduced, low variance, solved by a greedy bit-flip approach.

Key Benefits:

- Handles medium-variance columns more effectively than a strict “hard vs. easy” assumption.
- Presents a new complexity formula capturing partial enumerations and local flips.
- Potential synergy with ring-based LWE (RLWE) scenarios.

Preliminaries: LWE Definition

Search-LWE Problem:

$$\text{Given } \mathbf{A} \in \mathbb{Z}_q^{m \times n}, \quad \mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e} \pmod{q},$$

where:

- $\mathbf{s} \in \mathbb{Z}_q^n$ is the secret vector,
- $\mathbf{e} \in \mathbb{Z}_q^m$ is the error vector (small-norm distribution),
- m is the number of samples, n is dimension, q is the modulus.

Goal: Recover $\mathbf{s}$ given $\mathbf{A}$ and $\mathbf{b}$.

Sparse Binary Setting:

$$\mathbf{s} \in \{0, 1\}^n, \quad \text{Hamming weight} = h \ll n.$$

Preliminaries: Partial Lattice Reduction

The q -ary Embedding:

$$\Lambda = \begin{pmatrix} 0 & qI_n \\ \omega I_m & \mathbf{A} \end{pmatrix},$$

where ω is a *penalty parameter* controlling reduction vs. error amplification.

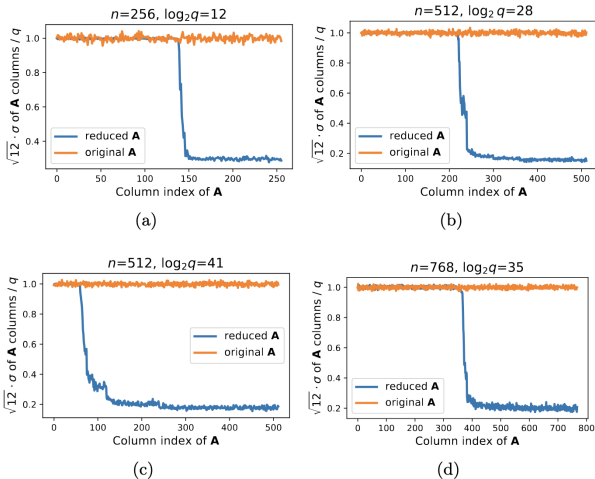
Result: After applying BKZ, flatter, etc., we get $\mathbf{A}' = \mathbf{R} \mathbf{A}$ and $\mathbf{b}' = \mathbf{R} \mathbf{b}$.

Measuring Reduction Quality:

$$\rho = \frac{\sigma(\mathbf{A}')}{\sigma(\mathbf{A}_{\text{init}})}, \quad \sigma(\mathbf{A}_{\text{init}}) = \frac{q}{\sqrt{12}}.$$

Small $\rho \implies$ better reduced columns, but bigger noise amplification.

Figure: Variance Separation



Conceptual Plot: The original $\mathbf{A}$ (orange) has uniform column variance $\approx q/\sqrt{12}$, while the reduced $\mathbf{A}'$ (blue) shows a significant drop after column index n_u .

Cruel vs. Cool Bits

Cruel bits:

- The n_u unreduced columns (remain $\sim$ uniform in $\mathbb{Z}_q$).
- Hard portion of the secret.

Cool bits:

- The $n_r = n - n_u$ reduced columns (low variance).
- Easy portion of the secret (recoverable with simple stats).

Key Observation: After partial reduction, the dot product $\mathbf{a} \cdot \mathbf{s}$ is dominated by the cruel bits if you guess them incorrectly. The correct guess yields a recognizable residual pattern, making it statistically distinguishable.

Cruel Bit Brute Force

Hardest step:

- We only guess $\mathbf{s}_{0:n_u}$, the cruel portion.
- Evaluate

$$\mathbf{RA} \cdot \mathbf{s}^* - \mathbf{Rb} \pmod{q}$$

for each guess $\mathbf{s}^*$ in that subspace.

- The *correct guess* exhibits narrower standard deviation or a distinct residual distribution (as shown in next figure).

Why feasible? n_u is significantly smaller than n ; or the attacker knows h_u (the smaller Hamming weight among the cruel bits).

Greedy Recovery of “Cool” Bits

Context: From original paper, once n_u “cruel” bits are correctly guessed, the remaining $n_r = n - n_u$ bits have lower variance, enabling a simple *bit-flip* approach.

Algorithm Greedy Recovery of Cool Bits

Require: $\mathbf{s}^*$ partially correct (cruel bits), the rest set to 0

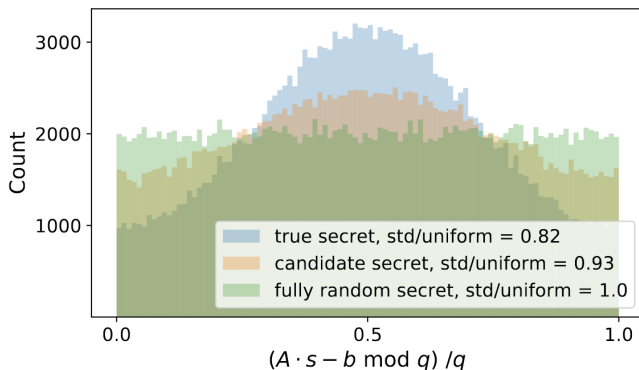
```

1: for  $i$  in  $\{n - n_r, \dots, n - 1\}$  do
2:    $s^*[i] \leftarrow 0$ 
3:    $\sigma_0 \leftarrow \sigma(\mathbf{A} \cdot \mathbf{s}^* - \mathbf{b} \bmod q)$ 
4:    $s^*[i] \leftarrow 1$ 
5:    $\sigma_1 \leftarrow \sigma(\mathbf{A} \cdot \mathbf{s}^* - \mathbf{b} \bmod q)$ 
6:   if  $\sigma_0 \leq \sigma_1$  then
7:      $s^*[i] \leftarrow 0$ 
8:   else
9:      $s^*[i] \leftarrow 1$ 

```

Statistical Representation

Residual Histogram Example:

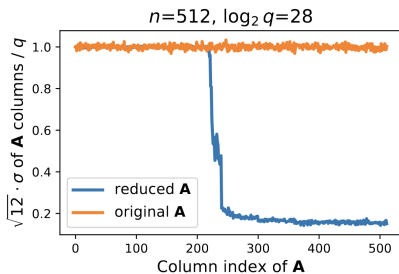
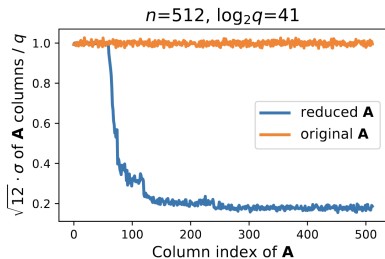


Blue: True secret guess, **Orange:** Partially correct guess, **Green:** Random guess.

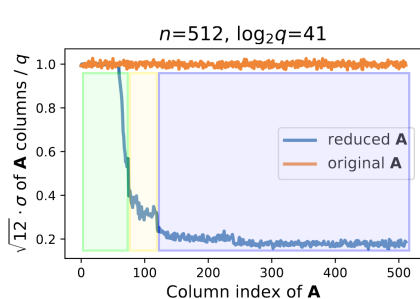
Variance Diagram Discussion

Empirical Observation:

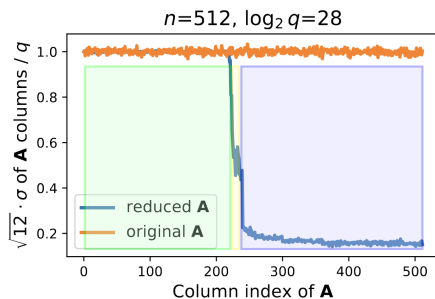
- After partial lattice reduction, columns sometimes exhibit a *gradual* drop in variance.
- A purely two-zone (Cruel vs. Cool) classification may fail if many columns have *moderate* variance.



Annotated Figures: Cruel, Chill, and Cool Regions



(a) $n = 512, \log_2 q = 41$.



(b) $n = 512, \log_2 q = 28$.

Figure: Division of columns into **Cruel** (green), **Chill** (yellow), and **Cool** (blue) zones, reflecting standard deviation changes post-reduction.

Proposed Method: Threshold-Based Zoning

Threshold Model:

Let σ_j be the standard deviation of column j in the reduced matrix $\mathbf{A}'$. Define two thresholds T_c and T_h such that

$$T_c > T_h > 0, \quad (\text{e.g., } T_c \approx 0.8 \sigma_u, \quad T_h \approx 0.3 \sigma_u).$$

We classify each column j by:

$$\text{Region}(j) = \begin{cases} \text{Cruel} & \text{if } \sigma_j \geq T_c, \\ \text{Chill} & \text{if } T_h \leq \sigma_j < T_c, \\ \text{Cool} & \text{if } \sigma_j < T_h. \end{cases}$$

Why This Works:

- $\sigma_j \approx q/\sqrt{12} \rightarrow$ treat as Cruel, requiring heavy enumeration.
- $\sigma_j \ll q/\sqrt{12} \rightarrow$ treat as Cool, a small variance for easy per-bit flips.
- σ_j in-between $\rightarrow$ use a *more refined* partial approach.

Why a Slope Might Appear

Partial Lattice Reduction in Practice:

- Techniques like `flatter` and BKZ may alternate or stall, producing columns that are *partially* reduced.
- This leads to some columns having **medium** variance, rather than a strict jump from $\approx q/\sqrt{12}$ to a low value.

Observation:

- Instead of a neat “cliff,” we see a *sloping* region in many empirical plots.
- Such semi-reduced columns are *neither* trivially brute forced *nor* trivially recovered by a single flip.

Implication: We need a more nuanced approach to handle these *intermediate* columns effectively.

Arguing for Three Zones: Cruel, Chill, & Cool

Our Three-Zone Decomposition:

$$\mathcal{I}_{\text{cruel}}, \mathcal{I}_{\text{chill}}, \mathcal{I}_{\text{cool}} \subseteq \{1, \dots, n\}, \quad \bigcup = \text{all columns.}$$

- **Cruel (Green):** High-variance columns needing brute force or partial guesses.
- **Chill (Yellow):** Transitional *sloping* columns, tackled with *probabilistic* or *ML-based* enumerations.
- **Cool (Blue):** Fully reduced columns, recovered by a simple **greedy flip**.

Motivation:

- *Reduced Overhead:* Limit brute force to truly uniform columns.
- *Structured Data:* Medium-variance columns can still be partially distinguished by advanced methods.
- *Adaptive Windowing:* Possibly shrink or expand Chill as needed.

Example Matrix Partition (I)

$$\begin{array}{c} \text{Cruel} \quad \text{Chill} \quad \text{Cool} \\ \left[\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & a_{26} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} & a_{36} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} & a_{46} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} & a_{56} \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} & a_{66} \end{array} \right] \end{array} \times \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \end{bmatrix}$$

Showcasing base matrix $\mathbf{A}$ with $n = 6$ example

Figure: Conceptual $n \times n$ matrix with columns labeled **Cruel** (green), **Chill** (yellow), and **Cool** (blue).

Note:

- We highlight columns in green/yellow/blue to illustrate partial-lattice reduction outcome.
- High-variance columns (Cruel) vs. transitional (Chill) vs. reduced (Cool).

Example Matrix Partition (II)

$$\begin{array}{c} \text{Cruel} \quad \text{Chill} \quad \text{Cool} \\ \left[\begin{array}{ccc|cc} a_{11} \cdot s_1 & a_{12} \cdot s_2 & a_{13} \cdot s_3 & a_{14} \cdot s_4 & a_{15} \cdot s_5 & a_{16} \cdot s_6 \\ a_{21} \cdot s_1 & a_{22} \cdot s_2 & a_{23} \cdot s_3 & a_{24} \cdot s_4 & a_{25} \cdot s_5 & a_{26} \cdot s_6 \\ a_{31} \cdot s_1 & a_{32} \cdot s_2 & a_{33} \cdot s_3 & a_{34} \cdot s_4 & a_{35} \cdot s_5 & a_{36} \cdot s_6 \\ a_{41} \cdot s_1 & a_{42} \cdot s_2 & a_{43} \cdot s_3 & a_{44} \cdot s_4 & a_{45} \cdot s_5 & a_{46} \cdot s_6 \\ a_{51} \cdot s_1 & a_{52} \cdot s_2 & a_{53} \cdot s_3 & a_{54} \cdot s_4 & a_{55} \cdot s_5 & a_{56} \cdot s_6 \\ a_{61} \cdot s_1 & a_{62} \cdot s_2 & a_{63} \cdot s_3 & a_{64} \cdot s_4 & a_{65} \cdot s_5 & a_{66} \cdot s_6 \end{array} \right] + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \end{bmatrix} \end{array}$$

Figure: Element-wise multiplication of $\mathbf{A}$ by $\mathbf{s}$ plus error $\mathbf{e}$, illustrating **Cruel**, **Chill**, and **Cool** columns.

Note:

- Each column remains colored based on variance classification.
- The final vector $\mathbf{b}$ is formed as $\mathbf{b} = \mathbf{As} + \mathbf{e}$.

Three Attack Phases

Phase 1: Brute Force (Cruel)

- Columns in $\mathcal{I}_{\text{cruel}}$ have variance $\approx \frac{q}{\sqrt{12}}$.
- Must guess bits, e.g., $\text{BruteForceCruel}(\mathbf{A}_{*, \mathcal{I}_{\text{cruel}}}, \mathbf{b})$.
- Let $\mathbf{s}_{\text{cruel}}$ be recovered cruel bits; $\text{cruel_ones} = \text{HW}(\mathbf{s}_{\text{cruel}})$.

Phase 2: Probabilistic (Chill)

- *Chill* region $\mathcal{I}_{\text{chill}}$ has moderate variance.
- Assign $p_j = \text{EstimateProb}(j, \mathbf{A}, \mathbf{s}_{\text{cruel}})$.
- Select $k = h - \text{cruel_ones}$ bits to be 1.
- Possibly refine guess with local flips if final check fails.

Phase 3: Greedy (Cool)

- $\mathcal{I}_{\text{cool}}$ columns: significantly reduced variance.
- Use **GreedyFlip** approach: flip bit if it lowers residual variance.

Multi-Zone Attack Algorithm

Algorithm: Cool, Chill, & Cruel

Algorithm Multi-Zone Recovery

Require: Reduced $\mathbf{A}$, vector $\mathbf{b}$, q , Hamming weight h

- 1: $(\mathcal{I}_{\text{cruel}}, \mathcal{I}_{\text{chill}}, \mathcal{I}_{\text{cool}}) \leftarrow \text{ClassifyColumns}(\mathbf{A}, T_c, T_h)$
 - 2: $\mathbf{s}_{\text{cruel}} \leftarrow \text{BruteForceCruel}(\mathbf{A}_{*, \mathcal{I}_{\text{cruel}}}, \mathbf{b})$
 - 3: $\text{cruel_ones} \leftarrow \text{HW}(\mathbf{s}_{\text{cruel}})$
 - 4: $p_j \leftarrow \text{EstimateProb}(j, \mathbf{A}, \mathbf{s}_{\text{cruel}}), j \in \mathcal{I}_{\text{chill}}$
 - 5: $k \leftarrow h - \text{cruel_ones}$
 - 6: $\mathbf{s}_{\text{chill}} \leftarrow \text{RankAndAssign}(p_j, k)$
 - 7: $\mathbf{s}_{\text{cool}} \leftarrow \text{GreedyFlip}(\mathbf{A}, \mathbf{b}, \mathbf{s}_{\text{cruel}}, \mathbf{s}_{\text{chill}})$
 - 8: $\text{RefineChill}(\mathbf{A}, \mathbf{b}, \mathbf{s}_{\text{cruel}}, \mathbf{s}_{\text{chill}}, \mathbf{s}_{\text{cool}})$
 - 9: **return** $(\mathbf{s}_{\text{cruel}}, \mathbf{s}_{\text{chill}}, \mathbf{s}_{\text{cool}})$
-

Key Points:

- ESTIMATEPROB uses variance or small ML models to rank bits in *Chill*.
- GREEDYFLIP is the standard bit-flip from earlier slides.
- REFINETCHILL tries local flips if the guess fails final checks.

Probability & Combinatorial Logic

Partition Counts:

- After partial-lattice reduction, columns split into $\mathcal{I}_{\text{cruel}}, \mathcal{I}_{\text{chill}}, \mathcal{I}_{\text{cool}}$.
- Let $n_{\text{cruel}}, n_{\text{chill}}, n_{\text{cool}}$ be the respective counts.
- Once the cruel bits are guessed, define

$$\text{cruel_1} = \text{HW}(\mathbf{s}_{\text{cruel}}), \quad k = h - \text{cruel_1}.$$

Here, k bits remain for Chill + Cool.

Probability-Based Ranking (Chill):

- For each column $j \in \mathcal{I}_{\text{chill}}$, assign $p_j = \Pr[s_j = 1]$.
- E.g., a heuristic $p_j \propto 1/\sigma_j$, or a small ML model, or a direct statistical test.
- Select $k_{\text{chill}} \leq k$ columns with highest p_j as 1; place the rest in $\mathcal{I}_{\text{cool}}$.

Local Search & Refinement

Local Search If Guess Is Wrong:

- Even with probability ranking, some columns may be misclassified.
- A small BFS or iterative flip approach can fix borderline columns without enumerating all $\binom{n_{\text{chill}}}{k_{\text{chill}}}$ subsets.

Example Strategy:

- **RefineChill:**
 - 1 Identify columns j with p_j near threshold.
 - 2 Try flipping a bit from 0 to 1 (or 1 to 0).
 - 3 Re-check residual distribution or variance $\sigma(\mathbf{A}\mathbf{s}^* - \mathbf{b})$.
 - 4 Keep flip if it improves correctness, discard otherwise.
- Repeat until no improvement or iteration limit reached.

Goal:

- Achieve high success rate on Chill bits without *full* enumeration.
- Balance overhead vs. accuracy in partial-lattice cryptanalysis.

Complexity: Overview and Formula

Overall Cost:

- *Cruel* brute force, e.g. $2^{n_{\text{cruel}}}$,
- *Chill* enumerations or partial BFS,
- *Cool* overhead, typically linear in n_{cool} .

Notional Complexity Formula:

$$\begin{aligned} \text{Complexity} &\approx 2^{n_{\text{cruel}}} + \binom{n_{\text{chill}}}{k_{\text{chill}}} \times \text{ML_overhead}(1) \times \left[1 + \text{liability_coef} \times (n - n_{\text{cruel}} - n_{\text{chill}}) \right]. \\ &= 2^{n_{\text{cruel}}} + \underbrace{\binom{n_{\text{chill}}}{k_{\text{chill}}} \times \text{ML_overhead}(1) \times \left[1 + \text{liability_coef} \times (\dots) \right]}_{\text{chill enumerations}} + \underbrace{(\dots)}_{\text{cool overhead}} \end{aligned}$$

(Schematic of the main terms: $2^{n_{\text{cruel}}}$, $\binom{n_{\text{chill}}}{k_{\text{chill}}}$, $\text{ML_overhead}(1)$, etc.)

Complexity: Explanation

Term-by-Term Explanation:

- 1 $2^{n_{\text{cruel}}}$: Exhaustive or MITM for high-variance columns.
- 2 $\binom{n_{\text{chill}}}{k_{\text{chill}}}$: Worst-case enumerations in Chill (often reduced by ranking/BFS).
- 3 $\text{ML_overhead}(1)$: Extra cost per candidate if using neural nets or advanced tests.
- 4 $\text{liability_coef} \times (\dots)$: Local flips or expansions upon guess failure.
- 5 **Cool overhead**: $O(n_{\text{cool}})$ from final bit-flip.

Comparison to Original Cool & Cruel Approach

Original Two-Zone Complexity:

$$2^{n_u} + O(n - n_u), \quad n_u \approx n_{\text{cruel}}.$$

- Columns are *either* fully brute forced or trivially flipped.
- Medium-variance columns can inflate n_u or cause repeated flips.

Our Three-Zone Method:

- Potentially lowers n_{cruel} by pushing some columns into the *Chill* set,
- Incur extra $\binom{n_{\text{chill}}}{k_{\text{chill}}}$ overhead but less brute force dimension overall.
- If the slope region is large, partial enumerations or ML can be more efficient than a huge 2^{n_u} .

Conclusion:

- *If* columns form a clear cliff, two-zone is simpler.
- *If* a slope is substantial, three-zone yields better scaling in practice.

Discussion: Three-Zone Approach

Three-Zone Motivation:

- Columns after reduction may not show a perfect “cliff” in variance.
- We introduce a **Chill** region for moderate-variance columns, bridging pure brute force (Cruel) and trivial bit-flip (Cool).

Method Highlights:

- *Cruel subset*: brute force or meet-in-the-middle.
- *Chill subset*: probabilistic or ML-driven ranking.
- *Cool subset*: final greedy bit-flip step.

Complexity Recap:

- Chill enumerations can reduce an otherwise large brute force dimension.
- Overhead arises from classification or local refinement (Section 7).

Balancing Lattice Reduction

Key Observation:

- Stronger reduction $\implies$ fewer cruel bits (n_{cruel}), but larger noise σ_e for Chill/Cool.
- We can tune penalty param ω and block sizes in BKZ/flatter to find an optimal middle ground.

Future Experiments Needed:

- Measure trade-off between enumerating more cruel bits vs. coping with bigger noise in Chill/Cool.
- Possibly adapt or incrementally choose ω mid-process if the slope region is large.

Future Work

1. Ternary/Gaussian Secrets

- Many HE systems use ternary or narrow Gaussian distributions.
- Chill-based classification needs adjusted probability models or ML training.

2. Combining with SALSA/Transformers

- SALSA, PICANTE, VERDE show ML can recover bits efficiently.
- Our Chill region could adopt such models, but overhead must remain manageable.

3. Real-World Library Impact

- HE libs (HElib, SEAL) use small/sparse secrets.
- Mitigations include randomizing secret distribution or sub-lattice rotations.

4. Parameter Tuning & Local Search

- Defining “liability coefficients,” iteration limits for flips in Chill remains open.
- Tuning approach to get practical speedups is crucial.

Conclusion

Cool, Chill, and Cruel Partition:

- Addresses limitations of a purely two-zone partial-lattice attack.
- Each subset (Cruel, Chill, Cool) uses a method suited to its variance regime.

Key Takeaways:

- The Chill region adds overhead but can dramatically reduce brute force size.
- Fine-tuning partial reduction (ω) and local search heuristics is essential.
- Empirical tests up to $n = 1024$ will confirm practicality for real HE systems.

Looking Ahead:

- Further synergy with ring-based LWE and advanced ML classification.
- Bridging classical cryptanalysis and modern ML to break small/sparse LWE secrets.

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